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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech (CBS/CSE)(CBCS) Degree Examinations

December 2024 - January 2025

22CBS121/23CSE126 – ETHICAL HACKING AND NETWORK DEFENSE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss the differences between Vulnerability Analysis and Penetration Testing.	10	L*2	1	1
b)	Discuss Red, Blue and Purple Teaming with their responsibilities.	10	L2	1	1
2. a)	What are the five key steps of Threat Modeling Process.? Discuss	10	L3	1	1
b)	Explain different phases of Ethical Hacking.	10	L2	1	1
3. a)	List and explain seven logical steps involved in Information gathering.	10	L2	2	1
b)	What are the different Vulnerability Assessment stages? Explain.	10	L2	2	1
Unit – II					
4. a)	Discuss MAC Flooding and DNS Spoofing.	10	L2	2	1
b)	List and explain types of Web Server Vulnerability.	10	L2	3	1
5. a)	What is Denial of Service? How DDoS attack works?	10	L2	3	1
b)	Explain web based password cracking Techniques.	10	L2	3	1
6. a)	Discuss Smurf and SYN flood Attacks in detail.	10	L2	4	1
b)	List and explain the most common purposes of a SQL injection attack.	10	L3	4	1
Unit – III					
7. a)	Discuss the following types of Firewalls				
	i) Stateless Firewalls	10	L2	4	1
	ii) Stateful Firewalls				
b)	Discuss Network Address Translation along with its types.	10	L2	5	1
8. a)	Discuss the following				
	i) Honeypots	10	L2	5	1
	ii) Honeynets				
b)	Discuss Virtual Private Network (VPN) along with its types.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
